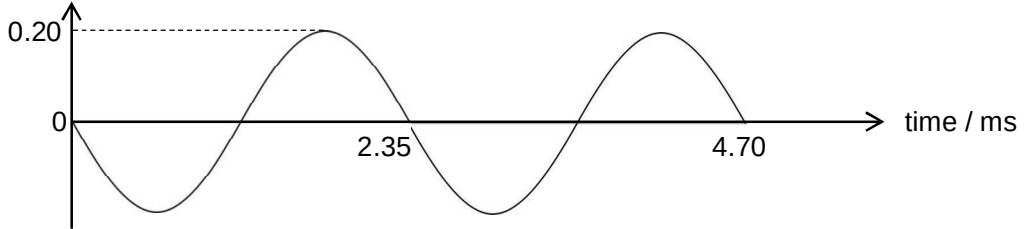


4a	From Fig. 4.1, wavelength $\lambda = 0.80 \text{ m}$ Using $v = f \lambda$ $f = v / \lambda = 340 / 0.8 = 425 \text{ Hz}$	C1 A1
4b	By COE, $E_T = \frac{1}{2} m v_{max}^2$ $v_{max} = a \omega = a 2\pi f$ $\therefore E_T = 2\pi^2 m f^2 a^2$	B1 B1 A0
4c	$E_T = 2\pi^2 m f^2 a^2$ $= 2\pi^2 \times 4.7 \times 10^{-26} \times 425^2 \times (0.2 \times 10^{-3})^2$ $= 6.7 \times 10^{-27} \text{ J}$	C1 A1

4d	displacement / mm  Sinusoidal wave with axes labelled (correct amplitude and period) Negative sine wave	B1 B1
4e	let θ be the phase angle between A and B. $0.10 = 0.20 \sin \theta$ $\theta = 30^\circ$	C1 A1

5a	When two waves meet at a point, the resultant displacement is equal to the	B1
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